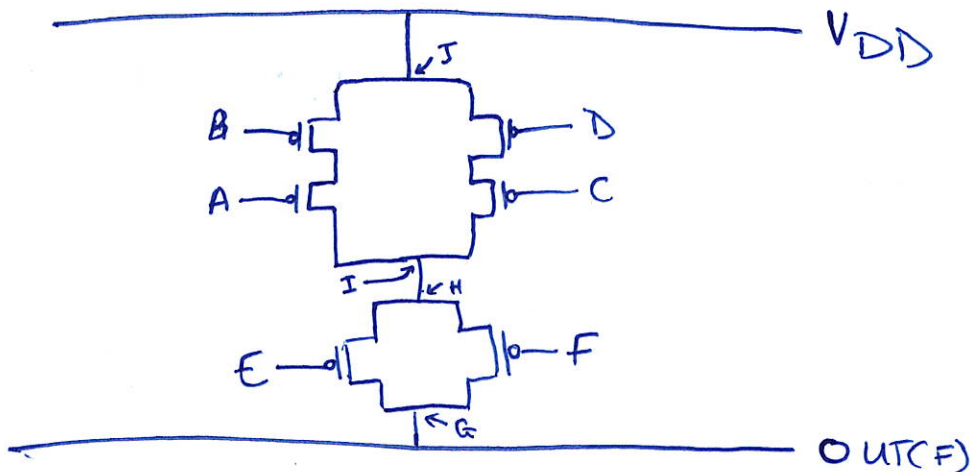


- (b) **Pull-up network.** Draw the PMOS pull-up network as the dual of the pull-down network. Clearly label all transistor gates and internal nodes. [4 points]



PMOS STRONG IS TOP. $F=1$

Notes on Notes Page 3

- (c) **Correctness check using input vectors.** Choose one input vector for which $F = 0$. Identify the conducting path from V_{out} to ground. Choose one input vector for which $F = 1$. Identify the conducting path from V_{DD} to V_{out} . [4 points]

$F=0$ when $A=0$ $B=0$ $C=0$ $D=0$ $E=1$ $F=1$

Path: $V_{out} \rightarrow T \rightarrow F_{\text{NMOS TRANSISTOR}} \rightarrow E_{\text{NMOS TRANSISTOR}} \rightarrow U \rightarrow \text{GROUND}$

$F=1$ when $A=1$ $B=1$ $C=0$ $D=0$ $E=1$ $F=0$.

Path: $V_{dd} \rightarrow J \rightarrow B_{\text{PMOS TRANSISTOR}} \rightarrow A_{\text{PMOS TRANSISTOR}} \rightarrow I \rightarrow H \rightarrow E_{\text{PMOS TRANSISTOR}}$

$\rightarrow G \rightarrow V_{out}$.

- (d) **Worst-case resistance before sizing.** Assuming all transistors are minimum size, identify the worst-case pull-down path resistance and the worst-case pull-up path resistance in terms of R_n . [4 points]

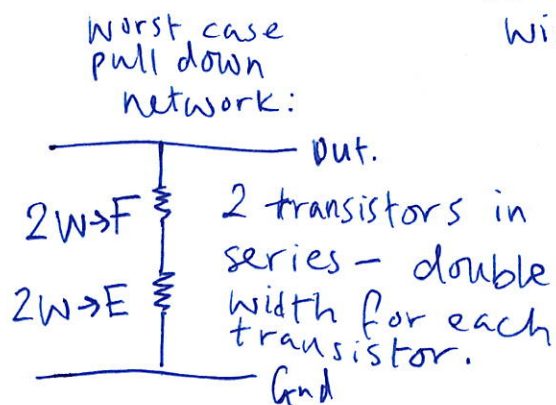
WORST CASE
PULL DOWN PATH:
 $V_{out} \rightarrow T \rightarrow F_{NMOS}$
 $\rightarrow E_{NMOS} \rightarrow U \rightarrow GND.$
2 NMOS TRANSISTORS W.
Resistance R_n :
WORST CASE PULL DOWN
PATH RESISTANCE = $2R_n$.

WORST CASE
PULL - UP PATH:
 $V_{out} \rightarrow G \rightarrow E_{PMOS} \rightarrow H \rightarrow I \rightarrow A_{PMOS} \rightarrow B_{PMOS} \rightarrow J \rightarrow VDD$
 $E \rightarrow A \rightarrow B$: 3 PMOS in series.
 $3R_p$, $R_p = 2R_n$.
Worst case Pull up
Path resistance = $6R_n$.

- (e) **Transistor sizing.** Choose integer transistor widths so that the worst-case pull-down resistance is approximately R_n , and the worst-case pull-up resistance is approximately $2R_n$. Report the chosen width of every transistor.

Your answer must include a short explanation of how your sizing changes the input capacitance seen by each signal.

[4 points]

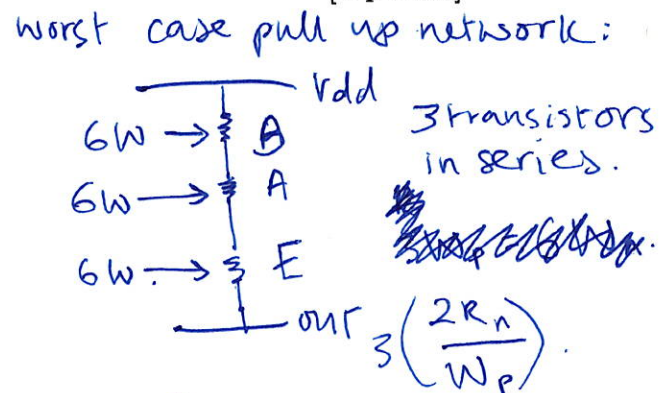


doubling
~~increasing~~ width increases
the input capacitance
(by 2x) because

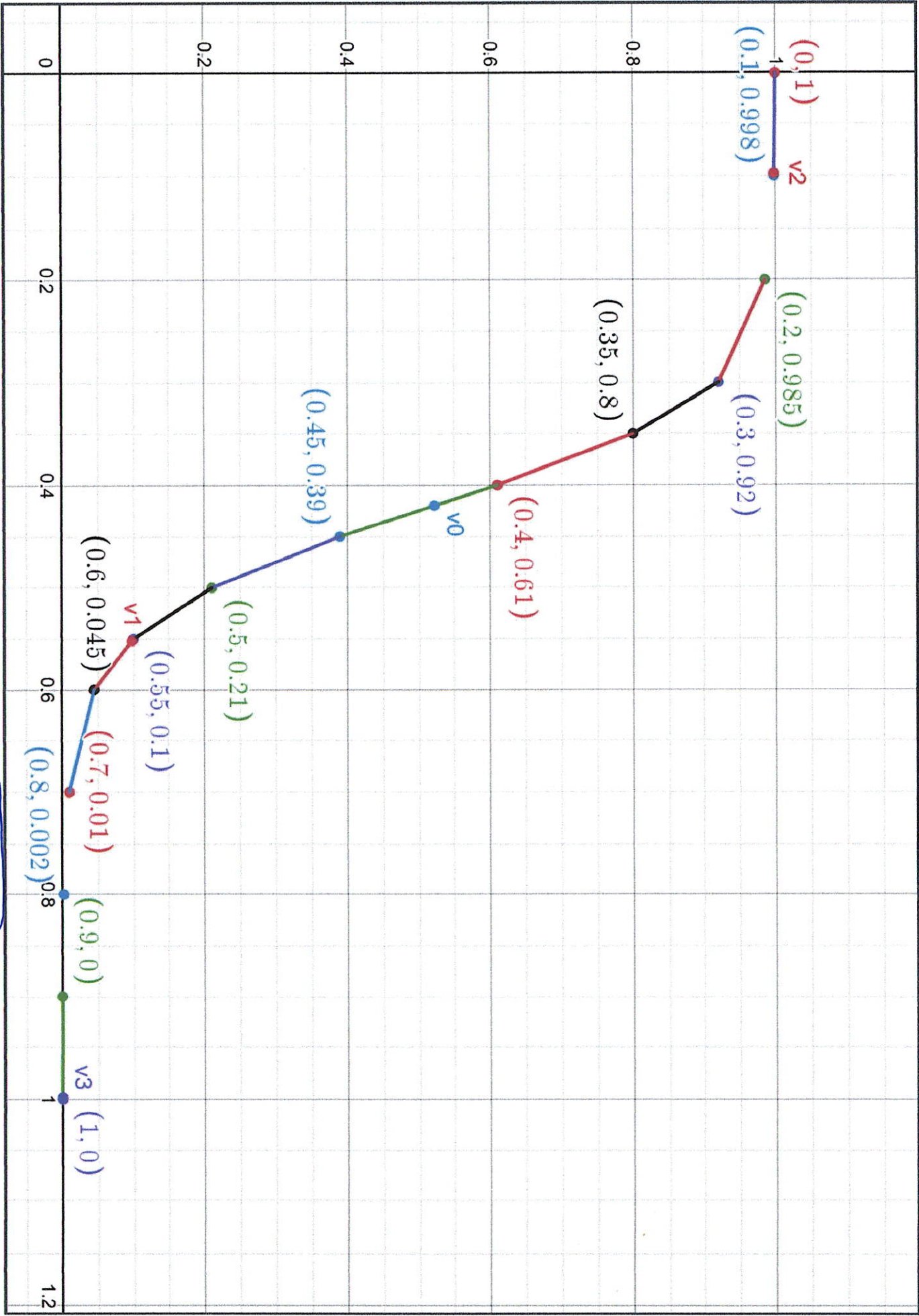
$$C_g \propto WL ?$$

↑
gate
capacitance

$W = \text{unit width}$



my increasing the
width of the ~~the~~ transistors
increases the input capacitance
seen by each
signal?



PRINTOUT 1

3 $(0.1, 0.998)$

4 $(0.2, 0.985)$

5 $(0.3, 0.92)$

6 $(0.35, 0.8)$

7 $(0.4, 0.61)$

8 $(0.45, 0.39)$

9 $(0.5, 0.21)$

10 $(0.55, 0.1)$

11 $(0.6, 0.045)$

12 $(0.7, 0.01)$

13 $(0.8, 0.002)$

14 $(0.9, 0)$

15 $(1, 0)$

16 approx curve

17 $y = -0.02x + 1 \{0 \leq x \leq 0.1\}$

18 $y = -0.65x + 1.115 \{0.2 \leq x \leq 0.3\}$

19 $y = -2.4x + 1.64 \{0.3 \leq x \leq 0.35\}$

20 $y = -3.8x + 2.13 \{0.35 \leq x \leq 0.4\}$

21 $y = -4.4x + 2.37 \{0.4 \leq x \leq 0.45\}$

22 $y = -3.6x + 2.01 \{0.45 \leq x \leq 0.5\}$

23 $y = -2.2x + 1.31 \{0.5 \leq x \leq 0.55\}$

24 $y = -1.1x + 0.705 \{0.55 \leq x \leq 0.6\}$

25 $y = -0.35x + 0.255 \{0.6 \leq x \leq 0.7\}$

26 $y = 0 \{0.9 \leq x \leq 1\}$

27 $\text{regenerating voltage inverters}$

Log in or sign up to save your beautiful math!

PRINTOUT 2

V_{in} 's range of $0.40 \rightarrow 0.45$ is wholly contained by the corresponding V_{out} values of $0.610 \rightarrow 0.390$.

(V_{in}, V_{out}) $(0.4, 0.610)$ $(0.45, 0.390)$. (x, y) .

~~XXXXXXXXXX~~ $m = \frac{0.39 - 0.61}{0.05} = -4.4$.

$$y = mx + c$$

$$c = 0.39 - (-4.4 \times 0.45) = 2.37.$$

$$V_m = -4.4(V_m) + 2.37. \text{ ie } V_{in} = V_{out}.$$

$$V_m + 4.4 V_m = 2.37$$

$$5.4 V_m = 2.37$$

$$V_m = \frac{2.37}{5.4} = 0.438 \bar{8} V$$

Notes Page 4
Q3A

Question 3: VTC, Noise Margins, and Regeneration [20 points]

A CMOS inverter has the following measured DC transfer characteristic. Assume:

$$V_{DD} = 1.0V.$$

V_{in} (V)	V_{out} (V)
0.00	1.000
0.10	0.998
0.20	0.985
0.30	0.920
0.35	0.800
0.40	0.610
0.45	0.390
0.50	0.210
0.55	0.100
0.60	0.045
0.70	0.010
0.80	0.002
0.90	0.000
1.00	0.000

FOR ENTIRE
QUESTION,
CHECKED
WORK AGAINST
GRAPH I MADE
ON DESMOS.
SHARED IN
PRINTOUTS 1&2.

Use only the data in the table. Do not assume the inverter is symmetric.

- (a) **Switching threshold.** Estimate the switching threshold V_M , defined as the point where

$$V_{in} = V_{out}.$$

Use linear interpolation between the two closest table entries.

[4 points]

See Notes Page 4
for working

$$V_M = 0.438 \text{ V}$$

$V_{in} \rightarrow V_{in}$

$$\frac{0.800 - 0.920}{0.05} = \frac{-0.120}{0.05} = -2.4 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.3 \rightarrow 0.35 \end{array}$$

$$\frac{0.610 - 0.8}{0.05} = -3.8 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.35 \rightarrow 0.4 \end{array}$$

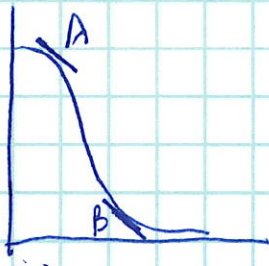
$$\frac{(0.390 - 0.610)}{0.05} = -4.4 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.4 \rightarrow 0.45 \end{array}$$

$$\frac{0.210 - 0.39}{0.05} = -3.6 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.45 \text{ to } 0.5 \end{array}$$

$$\frac{0.1 - 0.21}{0.05} = -2.2 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.5 \text{ to } 0.55 \end{array}$$

$$\frac{0.045 - 0.1}{0.05} = -1.1 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.55 \text{ to } 0.6 \end{array}$$

Notes Page 5
Q3B



Point A: Between V_{in} 0.2 and 0.35.
(V_{IL})

$M = -0.65 \rightarrow 1 \rightarrow M = -2.4$

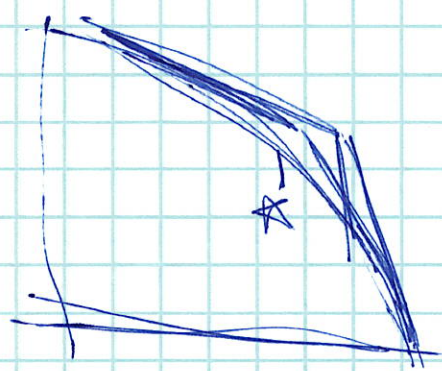
$M = -0.65$	$M = -0.65$	$M = -2.4$
0.2	0.25	0.3 0.35

$-2.4 + 0.65 \Rightarrow \Delta M = -1.75$

$-1 / -1.75 \approx 0.57$, which is less than $2/3$, so I can assume the point where

$m = -1$ for section A lies between $V_{in} = 0.2$ and $V_{in} = 0.3$. Furthermore,

~~0.2 + (1/1.75) * 0.15~~
= approx 0.28571



SIMILARLY, for POINT B: Between V_{in} 0.55 and 0.7
(V_{IH})

V_{in} 0.55 $\rightarrow$ V_{in} 0.7
 $-1.1 < 1 < -0.35$

$M = -1.1$	$M = -0.35$	$M = -0.35$
0.55	0.6 0.65	0.7

WRONG segments!

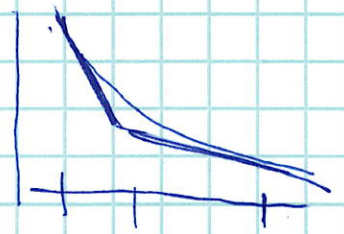
~~0.55 + (1/1.45) * 0.15~~

$M = -1.1$	$M = -0.35$	$M = -0.35$
0.55	0.6 0.65	0.7

$-0.35 \neq 1.1 \Rightarrow \Delta M = -1.45$

~~0.55 + (1/1.45) * 0.15~~ $\Rightarrow$ $-1 / -1.45 = 0.6897...$

$V_{IH} = 0.55 + \frac{1}{1.45} \times 0.15 = 0.653448...$



Notes Page 9
Q3c

(b) **Local slope estimation.** Compute the approximate slope

$$\frac{\Delta V_{out}}{\Delta V_{in}}$$

for each interval between $V_{in} = 0.30V$ and $V_{in} = 0.60V$. Identify the interval with the largest magnitude slope. [4 points]

The interval with the largest magnitude slope is $V_{in} (0.4 \rightarrow 0.45)$ with a slope of -4.4

Please see notes page 5.

(for approx. slopes for each interval & intermediate calculations)

(c) **Noise-margin estimation.** Estimate V_{IL} and V_{IH} using the condition

$$\frac{dV_{out}}{dV_{in}} = -1.$$

Because the data is discrete, you must explain which intervals you use to estimate V_{IL} and V_{IH} . Then compute

$$NM_L = V_{IL} - V_{OL}$$

and

$$NM_H = V_{OH} - V_{IH}.$$

Use $V_{OH} = 1.0V$ and $V_{OL} = 0V$.

[5 points]

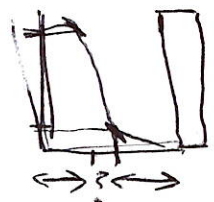
See Notes Page 9 for V_{IL} & V_{IH} calc.

$$V_{IL} = \text{approx } 0.28571$$

$$V_{IH} = \text{approx } 0.65345.$$

$$NM_L = V_{IL} - 0 = 0.28571V.$$

$$NM_H = V_{OH} - V_{IH} = 1 - V_{IH} = 0.34655V.$$



- (d) **Regeneration through a chain of inverters.** Suppose the input to the first inverter in a chain is

$$V_0 = 0.42V.$$

Use interpolation from the table to estimate

$$V_1, V_2, V_3,$$

where V_i is the output of the i -th inverter. Based on these values, explain whether the signal regenerates toward a legal logic value. [5 points]

$$V_{0\text{in}} = 0.42. \quad V_{0\text{-out}} = 0.552 \quad (y = -4.4x + 2.37)$$

$$V_{1\text{in}} = 0.552 \quad V_{1\text{-out}} = 0.0978 \quad (y = -1.1x + 0.705)$$

$$V_{2\text{in}} = 0.0978 \quad V_{2\text{-out}} = 0.998044 \quad (y = -0.02x + 1)$$

$$V_{3\text{in}} = 0.998044 \quad V_{3\text{-out}} = 0 \quad (y = 0.)$$

the signal oscillates closer and closer to 0 and 1 (magnitude of the) but not towards a legal logical value it depends on the number of inverters in the chain?

USING M values from Q3B, Pg 5 notes.

- (e) **Design interpretation.** If this inverter were used in a noisy environment, would you prefer a steeper or shallower VTC transition region? Explain your answer in terms of noise margins and regeneration. [2 points]

I would prefer a steeper transition region because I would want to reduce the range of voltages that would regenerate poorly & would want to maximise the noise margins by increasing the 'allowable' ranges of voltages.

